

After obtaining the quality label from human annotators, we apply bootstrapping with 10,000 resamples to estimate the equal opportunity self-preference bias. The annotation interface and instructions can be found in Appendix D and Appendix E.

4. Data and Experimental Design

To empirically examine AI self-preference bias as defined in Section 3, we design a series of resume correspondence experiments (Bertrand and Mullainathan 2004) in which LLMs act as evaluators (evaluator LLM) tasked with screening and selecting between pairs of resumes. In each pair, one resume is generated by the evaluator LLM itself, while the other is written by a human or produced by an alternative LLM. This setup allows us to test whether an evaluator LLM systematically favors its own outputs over others.

To construct these resume pairs, we begin with a dataset of human-written resumes. For each human-written resume, we use several state-of-the-art LLMs to generate counterfactual versions that describe the same candidate profile. We then form multiple types of resume pairs: evaluator LLM-generated versus human-written, and evaluator LLM-generated versus alternative LLM-generated. These pairs serve as the input for the evaluator LLMs.

In the subsections that follow, we first describe the original resume dataset. We then outline our experimental procedures, including the generation of counterfactual resumes and the design of the pairwise comparisons.

4.1. Data

We draw on a publicly available dataset containing 2,484 anonymized, human-written resumes collected from the professional resume-building platform LiveCareer.com (Bhawal 2021). The resumes were authored by real job seekers prior to the widespread adoption of large language models, ensuring that the content reflects human-written resumes rather than AI-generated text and thus making it well-suited for our study of AI self-preferencing. The dataset has been widely used in recent research on AI and algorithmic hiring. For example, it has served as a benchmark for evaluating hiring biases in language models (Wang et al. 2024), and has been used to investigate gender, racial, and intersectional biases in resume screening using language model retrieval techniques (Wilson and Caliskan 2024).

Spanning 24 distinct occupational categories, including teachers, consultants, chefs, engineers, and more, the dataset offers a diverse representation of professional backgrounds. Each resume typically includes multiple sections: an executive summary, education, work experience, and skills. Among these, the executive summary is particularly relevant to our study. This section comprises a free-text narrative in which candidates synthesize their qualifications, accomplishments, and career objectives, and is often used by evaluators to form initial impressions. Unlike structured sections

Table 1 Summary Statistics of Resume Summaries

Measure	Mean	Std. Dev.	Min	1st Quartile	Median	3rd Quartile	Max
Number of words	70.74	72.05	3.00	32.00	50.00	84.00	1216.00
Number of sentences	3.85	4.11	1.00	2.00	3.00	5.00	61.00
Average words per sentence	21.78	17.33	1.50	14.00	18.00	24.00	369.00
Number of unique words	52.53	39.32	2.00	28.00	42.00	64.00	520.00
Type-Token Ratio	0.82	0.10	0.32	0.75	0.83	0.89	1.00
Presence of numbers	0.37	0.48	0.00	0.00	0.00	1.00	1.00

Notes: The Type-Token Ratio (TTR) is a commonly used measure of lexical diversity, calculated as the number of unique words (types) divided by the total number of words (tokens) in a text.

such as education or work experience, which largely report verifiable facts, the executive summary is inherently subjective and stylistically flexible. As a result, it offers substantial scope for variation in phrasing, tone, and presentation—dimensions along which LLMs may exert influence. This makes the executive summary a natural and empirically relevant focus for examining how AI-generated text shapes perceptions and potentially affects hiring outcomes.

Thus, we focus our analysis on the executive summary section of each resume. To systematically construct counterfactual resumes, we replace the original executive summary of each human-written resume with an LLM-generated version, while preserving all other content (e.g., work experience, skills, education) unchanged. This approach isolates the effect of LLM-generated text on candidate evaluation outcomes, holding constant the factual and objective components of the resume. By doing so, we avoid potential confounds introduced by allowing LLM to modify structured information, which could lead to hallucinations or factual inaccuracies.

This dataset forms the basis of our experimental design. To construct a clean and comparable set of resumes, we preprocess the data by extracting the executive summary sections, removing formatting artifacts, and excluding observations with empty or missing summaries. After cleaning and validation, the final sample consists of 2,245 resumes, which are used in our correspondence experiments. Table 1 reports descriptive statistics for the human-written executive summaries. These summaries exhibit substantial variation in word and sentence counts, reflecting heterogeneity in writing styles across resumes. Measures of lexical diversity, including the type-token ratio, further indicate meaningful variation in vocabulary usage across summaries.³

4.2. Experimental Design

To examine AI self-preference bias in the context of algorithmic hiring, we design a series of resume correspondence experiments comprising two steps: counterfactual resume generation and pairwise resume evaluation, as illustrated in the Figure 2.

³ While there is no universally accepted threshold for high lexical diversity or an optimal type-token ratio, values above 0.5 are often considered indicative of high lexical diversity in short texts.

4.2.1. Counterfactual Resume Generation

We evaluate a diverse set of large language models that vary along two dimensions relevant to real-world hiring and resume-writing contexts: access regime (closed- versus open-source) and model scale. Specifically, we consider three closed-source models (GPT-4o, GPT-4o-mini, and GPT-4-turbo) and four open-source models (LLaMA 3.3-70B,⁴ Mistral-7B, Qwen 2.5-72B, and DeepSeek-V3). These models represent widely used and competitive systems for text generation and summarization tasks and exhibit broadly comparable performance on standard summarization benchmarks, as reported by the ProLLM leaderboard (ProLLM Team 2024). Our analysis does not rely on exact parity in benchmark performance; rather, these models capture a relevant cross-section of contemporary LLMs that job seekers or platforms may plausibly employ in resume preparation. To examine how model scale relates to self-preferencing behavior, we additionally include two smaller-scale language models—LLaMA 3.2-1B and LLaMA 3.2-3B. These models are not intended to represent tools commonly used in hiring contexts, but instead allow us to explore whether self-preference varies systematically with model capacity.

To generate resume summaries, we prompt each LLM with a modified version of the original resume in which the human-written summary is removed, but all other sections (e.g., work experience, skills, education) are left unchanged. The prompted LLM then generates a new resume summary, which is inserted back into the original resume to form a complete counterfactual version. By construction, each counterfactual resume differs from the original only in the linguistic expression of the executive summary, holding constant all underlying candidate attributes. The full prompt used for summary generation is provided in Appendix B.

To control for verbosity bias (Saito et al. 2023), where LLM evaluators may favor longer, verbose responses, we explicitly instruct each LLM to generate summaries within a specified word range, corresponding to the 1st and 3rd quartiles of the empirical length distribution of human-written summaries. This constraint minimizes variation in length across models and prevents length from confounding downstream evaluations.

The generation process described above is illustrated in Figure 2a. For each original human-written resume, each LLM produces a corresponding counterfactual version that differs only in the executive summary section. All other resume components are held fixed. This design isolates the effect of the *source of the summary text*—whether it is human-written, generated by the evaluator LLM, or generated by an alternative LLM. In the next subsection, we describe how these counterfactual resumes are paired and evaluated to systematically test for AI self-preference bias.

⁴ The suffix (e.g., “70B”) refers to the number of parameters in the model—in this case, 70 billion. Larger models generally have greater capacity for language understanding and generation, though size alone does not determine overall performance.